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#include <stdio.h>

#define MAX 10

int a[MAX], n = 0;

void display()
{
    int i;
    printf("List: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

void insert()
{
    int pos, value, i;

    if (n == MAX) {
        printf("List is full\n");
        return;
    }

    printf("Enter position and value: ");
    scanf("%d %d", &pos, &value);

    if (pos < 1 || pos > n + 1) {
        printf("Invalid position\n");
        return;
    }

    for (i = n; i >= pos; i--)
        a[i] = a[i - 1];

    a[pos - 1] = value;
    n++;

    printf("After insertion: ");
    display();
}

void deleteElement()
{
    int pos, i;

    if (n == 0) {
        printf("List is empty\n");
        return;
    }

    printf("Enter position to delete: ");
    scanf("%d", &pos);

    if (pos < 1 || pos > n) {
        printf("Invalid position\n");
        return;
    }
}

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        for (i = pos - 1; i < n - 1; i++)
            a[i] = a[i + 1];

        n--;

        printf("After deletion: ");
        display();
    }

void search()
{
    int value, i, found = 0;

    printf("Enter element to search: ");
    scanf("%d", &value);

    for (i = 0; i < n; i++) {
        if (a[i] == value) {
            printf("Element found at position %d\n", i + 1);
            found = 1;
            break;
        }
    }

    if (!found)
        printf("Element not found\n");

    printf("After search: ");
    display();
}

int main()
{
    int choice;

    do {
        printf("\n--- ARRAY LIST ---\n");
        printf("1. Insert\n");
        printf("2. Delete\n");
        printf("3. Search\n");
        printf("4. Display\n");
        printf("5. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                insert();
                break;
            case 2:
                deleteElement();
                break;
            case 3:
                search();
                break;
            case 4:
                display();

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        break;
    case 5:
        printf("Exit\n");
        break;
    default:
        printf("Invalid choice\n");
    }
} while (choice != 5);

return 0;
}
```

Output:

```
--- ARRAY LIST ---
1. Insert
2. Delete
3. Search
4. Display
5. Exit
Enter choice: 1
Enter position and value: 1 10
After insertion: List: 10
```